

RMO– 2015(4)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions. All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABC be a triangle. Let B' denote the reflection of B in the internal angle bisector ℓ of $\angle A$. Show that the circumcentre of the triangle $CB'I$ lies on the line ℓ , where I is the incentre of ABC .
 2. Let $P(x) = x^2 + ax + b$ be quadratic polynomial with a is real and b is rational. Suppose there are real numbers. Suppose such that $\langle P(0)^2, P(1)^2, P(2)^2 \rangle$ are integers. Prove that a and b are integers.
 3. Find all integers a, b, c such that $a^2 = bc + 4, \quad b^2 = ca + 4$.
 4. Suppose 40 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
 5. Two circles Γ and Σ in the plane intersect at two distinct points A and B . A line through B intersects Γ and Σ again in C and D , respectively. Suppose $CA = CD$. Show that the centre of Σ lies on Γ .
 6. How many integers m satisfies both the properties:
 - (i) $1 \leq m \leq 5000$,
 - (ii) $[\sqrt{m}] = [\sqrt{m+125}]$?
 (Here $[x]$ denotes the largest integer not exceeding x , for any real number x .)

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